A Decision Support System in Single-Supplier, Multi-Buyer Supplier-Managed Inventory: A Case Study in the Automotive Industry

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Abstract. This study proposes a mixed-integer linear programming model to address the single-supplier, multi-buyer supplier-managed inventory problem by optimizing order allocation under capacity and inventory constraints. A decision support system is developed to integrate enterprise data, model instantiation, and automated solution generation within an interactive framework. The approach is validated through a real case study in the automotive industry. Compared with the baseline allocation policy, the optimized solution reduces total system cost by 95.41%, demonstrating substantial economic benefits from coordinated capacity allocation. Results show that a small number of severe shortages dominate total cost, while the model produces stable and targeted capacity allocation decisions with minimal violations. The proposed framework provides a structured foundation for integrating optimization into practical supply chain planning and offers flexibility for future extensions.

Keywords: Decision Support System, Supplier-Managed Inventory, Inventory Management

1 Introduction

Vertical integration has been extensively investigated as a key corporate strategy in the fields of strategic management and organizational economics [1]. In the context of supply chain management, vertical integration refers to a firm's ownership and control over multiple stages of the value chain and, therefore, can be considered an attractive approach to supply chain design and reconfiguration [2]. However, when implementing vertical integration in their supply chains, firms must carefully determine whether to focus on upstream activities related to raw materials or downstream activities that convert these materials into final products [3].

Regarding upstream vertical integration, vendor-managed inventory (VMI) can shift inventories toward upstream stages of the supply chain [4]. VMI is a partnering initiative aimed at enhancing efficiency across supply chains by allowing the supplier to monitor the buyer's inventory and make replenishment decisions regarding order quantity, timing, and delivery [5, 6]. Under the VMI program, the vendors' responsibility in managing inventory has been emphasized [7]. Therefore, effective VMI implementation requires the sharing of information among supply chain partners [8]. Based on VMI

principles, Supplier-managed inventory (SMI) is a similar approach but focuses on the inbound flow to manufacturing, where material replenishment is triggered by the manufacturer's production schedule, rather than restocking finished goods [9]. To the best of our knowledge, SMI has received considerably less attention than VMI in existing literature. Optimization-oriented SMI model remains relatively limited and underdeveloped compared to the extensive research on VMI.

This study considers a Single-Supplier, Multi-Buyer Supplier-Managed Inventory problem (SS-MB-SMI) through a case study in the automotive supply chain. In this context, the supplier manufactures and provides component sets for car seat covers, which are then used by the buyer for vehicle assembly. Both companies belong to the same corporate group, reflecting a vertical integration strategy within the supply chain. This setup enables SMI implementation to allow the supplier to monitor the buyers' component inventory levels and allocate the purchased order to balance supply among them. The problem is then formulated as a mixed-integer linear programming (MILP) model to allocate orders among buyers based on capacitated production constraints. Finally, a decision support system (DSS) is developed to assist the supplier in managing inventory, allocating orders, and balancing supply among multiple buyers, demonstrating the practical application of the proposed SMI model.

The following section of this article is structured as follows: Sect. 2 reviews the relevant literature and clarifies the research contribution. Sect. 3 presents the mathematical formulation of the SS-MB-SMI model. Sect. 4 describes the decision support system framework. Sect. 5 reports the results and evaluation. Finally, Section 6 concludes the study by summarizing the main findings and directions for future research.

2 Literature review

VMI has attracted considerable attention as an effective mechanism for coordinating supply chain activities of single-vendor multi-buyer models [10]. To the best of our knowledge, although SMI is closely related to VMI, there are almost no studies that propose mathematical models for SMI. For this reason, Table 1 reviews existing VMI-based studies to provide a reference point, thereby clarifying the research gap in SMI that this study aims to address.

Research on single-vendor multi-buyer inventory systems has been extensively developed within the VMI paradigm, with the primary objective of improving coordination between vendors and buyers through optimized replenishment decisions. Early studies focused on deterministic demand, mainly to demonstrate the efficiency gains achieved through centralized coordination compared to decentralized optimization [11, 12].

Later research improved their models by considering more realistic operational factors, including multiple products, capacitated production, and alternative stockout responses such as backorder and lost sales [13, 14, 15]. Despite these extensions, most papers largely remain within the VMI framework, where replenishment decisions are optimized to minimize total cost, emphasizing cost-minimizing replenishment policies,

with limited attention to how suppliers actively allocate constrained capacity among multiple buyers.

More recent studies have introduced industry-specific characteristics to better reflect real-world inventory dynamics. For instance, the authors develop a single-vendor multi-buyer VMI model with multiple quality-degraded raw materials in the agrifood industry and apply Whale Optimization Algorithm (WOA) to maximize total profit [16]. Although this work significantly enhances the practical relevance of the model, it mainly focuses on algorithmic performance and inventory dynamics, rather than providing structured decision support at the system level.

In parallel, several studies have employed metaheuristic and fuzzy-based approaches to address large-scale and uncertain environments [17, 18]. However, these contributions generally lack an explicitly defined Decision Support System (DSS) architecture that links optimization outputs to managerial decision processes in semi-structured contexts.

Overall, the literature reveals three interrelated reseach gaps. First, mathematical optimization models for SMI in single-supplier multi-buyer settings are still limited. Second, capacity allocation is rarely modeled as a core decision variable within such models. Third, the integration of optimization models into a DSS framework supporting what-if scenarios and sensitivity analysis remains underexplored. Addressing these gaps, the present study develops a MILP-based SS-MB-SMI model and integrates it into a DSS that supports managerial evaluation of trade-offs among capacity, inventory policies, and service performance.

This study makes three main contributions. First, it provides a formal MILP formulation for SMI in a single-supplier, multi-buyer context, extending the limited optimization literature on SMI. Second, it considers capacity allocation as a strategic decision variable, linking production constraints with inventory coordination. Third, by embedding the model into a DSS framework, the study offers a practical tool that enables scenario analysis and managerial decision support, bridging the gap between optimization theory and operational practice.

Table 1. Related work

Article	Problem characteristics							Solution Method	
	Industry*	Replenishment Programs**	Multiple products	Production Capacity	Order multiple	Overstock penalty	Stockout response***	Modelling and/or Algorithm****	DSS
[11]	_	V	_	_	X	_	_	NLP, SS	_
[12]	_	V	_	_	X	X	_	NLP, EXA	_
[13]	_	V	X	X	X	_	B & L	NLP, PSO, GA	_
[14]	_	V	_	X	X	X	B	NLP, EXA	_

[15]	_	V	X	X	_	_	B & L	NLP, LR	_
[16]	F	V	_	X	_	_	_	NLP, WOA	_
[17]	_	V	_	_	_	_	L	Fuzzy	_
[18]	_	V	_	X	_	_	B & L	NLP, PSO	_
[19]	F	V	_	X	_	_	_	NLP, EXA	_
[20]	_	V	_	_	_	X	_	NLP, HEU	_
[21]	F	V	_	X	_	_	L	GA	Y
[22]	_	V	_	_	X	_	_	GA	_
This paper	A	S	X	X	X	X	B	MILP	Y

* A: Automotive, F: Food

** V: VMI, S: SMI

*** B: Backorder, L: Lost sales

**** EXA: Exact approaches, GA: Genetic Algorithm, HEU: Heuristic, NLP: Non-linear Programming, MILP: Mixed Integer Linear Programming, LR: Lagrangian Relaxation, PSO: Particle Swarm Optimization, SSM: Sequential Solution, WOA: Whale Optimization Algorithm

3 Problem definition and formulation

This paper examines an SS-MB-SMI for a component supply chain including two firms within a vertically integrated corporate structure, where the upstream firm produces components and supplies to the downstream firm for final product manufacturing (Fig 1). In this problem, there is a single component supplier (CS) which provides I components for J finished-goods plants (FGP) over a planning horizon of T periods (Figure 1). Each FGP has a specified case-pack requirement $CP_{i,j}$ for each item, which indicates the number of units needed to be packed in a single shipping box to minimize wasted space and optimize logistics efficiency. At each period, the production capacity $CAP_{i,t}$ of each item is predefined, reflecting the manufacturing capabilities of the CS. The safety stock is determined based on the inventory ceiling $U_{i,j,t}$ and floor $L_{i,j,t}$, aiming to ensure demand fulfillment under uncertainty and to minimize the risk of supply chain disruptions.

The inventory level of components at each FGP is denoted as $BI_{i,j}$ at the beginning and changes over time by a quantity $\Delta I_{i,j,t}$ due to demand fluctuation and/or transshipments between warehouses. Regarding cost, the cost components can be detailed as follows:

- Backorder cost: arise when components are unavailable at FGPs, causing production shortages and delaying manufacturing to a later period.
- Overstock cost: occur when net inventory at FGPs exceeds the inventory ceiling $U_{i,j,t}$, including extra warehouse space, handling, and potential obsolescence costs.
- Shortage cost: reflects the impact of net inventory falling below the inventory floor $L_{i,j,t}$, increasing the risk of stockouts under demand uncertainty.

- Penalty cost: result from shipments that do not meet case-pack requirements, causing inefficient box utilization

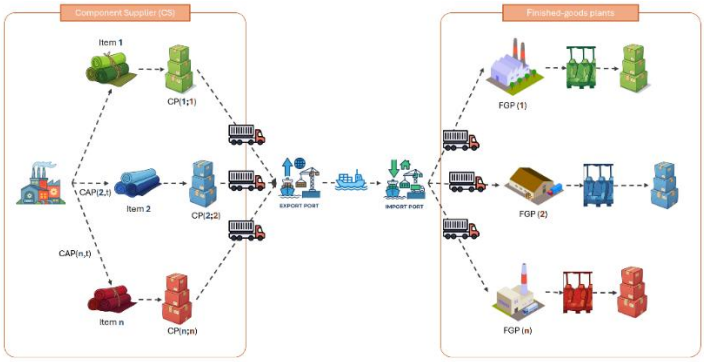

Fig. 1. The component supply chain

3.1 Assumptions and notations

The following assumptions are made:

- Input parameters are deterministic
- Stockouts are allowed to be fulfilled as backorders
- The supplier's production capacity for each item is predetermined and fixed for each period. Accordingly, the quantity of firmed purchase orders is equal to this capacity in each period.
- The shipment lead time from the CS to FGP is constant. Since the CS manages the inventory in advance to cover this fixed lead time, the shipping lead time can therefore be ignored.

The notations used in the proposed model can be listed as Table 2.

Table 2. Indices, input parameters, and variables

Indices	
$\boldsymbol{i}$	Items, $i = (1,2,\dots,I)$
$\boldsymbol{j}$	Finished-goods plants, $j = (1,2,\dots,J)$
$\boldsymbol{t}$	Time periods, $t = (1,2,\dots,T)$
Parameter	
$\boldsymbol{BI_{i,j}}$	Beginning inventory of item i at finished-goods plant j
$\boldsymbol{CP_{i,j}}$	Case-pack size of item i at finished-goods plant j
$\boldsymbol{C^b_{i,j,t}}$	Backorder unit cost of item i at finished-goods plant j at period t
$\boldsymbol{C^o_{i,j,t}}$	Overstock unit cost of item i at finished-goods plant j at period t
$\boldsymbol{C^s_{i,j,t}}$	Shortage unit cost of item i at finished-goods plant j at period t
$\boldsymbol{C^p_{i,j,t}}$	Penalty cost of item i at finished-goods plant j at period t

$\boldsymbol{U_{i,j,t}}$	Inventory ceiling of item i at finished-goods plant j at period t
$\boldsymbol{L_{i,j,t}}$	Inventory floor of item i at finished-goods plant j at period t
$\boldsymbol{CAP_{i,t}}$	Production capacity of item i at period t
$\Delta\boldsymbol{I_{i,j,t}}$	Inventory fluctuation of item i at finished-goods plant j at period t
$\boldsymbol{HV}$	High value
Variables	
$\boldsymbol{q_{i,j,t}}$	Number of full case-packs of item i firmed by finished-goods plant j at period t
$\boldsymbol{r_{i,j,t}}$	Residual units of item i firmed by finished-goods plant j at period t
$\boldsymbol{I_{i,j,t}}$	Net inventory of item i at finished-goods plant j at period t
$\boldsymbol{bo_{i,j,t}}$	Backorder units of item i at finished-goods plant j at period t
$\boldsymbol{o_{i,j,t}}$	Overstock units of item i at finished-goods plant j at period t
$\boldsymbol{s_{i,j,t}}$	Shortage units of item i at finished-goods plant j at period t
$\boldsymbol{p_{i,j,t}}$	Binary variable which is equal to 1 if the firm POs of item i at finished-goods plant j at period t are not divisible by the case-pack size. Otherwise, it is equal to 0.

3.2 Mathematical model

The developed mixed integer linear programming model for the SS-MB-SMI problem can be formulated as follows:

$$Min\left[\sum_i^I \sum_j^J \sum_t^T \left(C_{i,j,t}^o \times o_{i,j,t} + C_{i,j,t}^s \times s_{i,j,t} + C_{i,j,t}^b \times bo_{j,j,t} + C_{i,j,t}^p \times p_{i,j,t}\right)\right] \quad (1)$$

$$I_{i,j,1} = BI_{i,j} + \Delta I_{i,j,1} + q_{i,j,1} \times CP_{i,j} + r_{i,j,1},\ \forall i, j \quad (2)$$

$$I_{i,j,t} = I_{i,j,(t-1)} + \Delta I_{i,j,t} + q_{i,j,t} \times CP_{i,j} + r_{i,j,t},\ \forall i, j, t \geq 2 \quad (3)$$

$$\sum_j^J \left(q_{i,j,t} \times CP_{i,j} + r_{i,j,t}\right) = CAP_{i,t},\ \forall i, t \quad (4)$$

$$bo_{i,j,t} \geq -I_{i,j,t},\ \forall i, j, t \quad (5)$$

$$o_{i,j,t} \geq I_{i,j,t} - U_{i,j,t},\ \forall i, j, t \quad (6)$$

$$s_{i,j,t} \geq L_{i,j,t} - I_{i,j,t},\ \forall i, j, t \quad (7)$$

$$r_{i,j,t} \leq HV \times p_{i,j,t},\ \forall i, j, t \quad (8)$$

$$r_{i,j,t} \geq HV \times \left(p_{i,j,t} - 1\right) + 1,\ \forall i, j, t \quad (9)$$

$$bo_{i,j,t}, o_{i,j,t}, s_{i,j,t} \geq 0,\ \forall i, j, t \quad (10)$$

$$q_{i,j,t}, r_{i,j,t} \in N,\ \forall i, j, t \quad (11)$$

$$p_{i,j,t} \in \{0; 1\},\ \forall i, j, t \quad (12)$$

The objective (1) minimizes the total cost, including backorder, overstock, shortage, and penalty cost. Constraints (2) and (3) define the inventory net at time periods. Constraint (4) ensures that the total firm purchased orders at any period must be equal to the predefined capacity. Constraints (5)-(7) determine the backorder, overstock, and understock quantity, respectively. Constraints (8) and (9) linearize the binary penalty variable. Constraints (10)-(12) present conditions of variables.

4 Decision support system framework

The Decision Support System (DSS) developed in this study is designed to support semi-structured decision problems in an SMI context, where part of the decision-making process can be formally modeled and optimized, while the remaining part still requires the assessment and judgment of the planner. In this setting, the DSS is not intended to replace the planner, but rather to provide decision support by enabling the integration of structurally defined decision components with elements that depend on situational evaluation [23].

4.1 System architecture

The system architecture integrates enterprise data, optimization modeling, and managerial evaluation within a unified, layered structure (Fig 2). Rather than operating as a standalone optimization engine, the DSS connects data processing, model execution, and user interaction in a continuous decision loop.

Data Source Layer

The architecture begins with operational processes that generate raw transactional data related to production, inventory, and distribution. These data form the primary input to the analytical system.

Data Warehouse Layer

Raw data are processed through an ETL procedure and stored in the Normalized Data Store (NDS), where consistency, completeness, and temporal alignment are ensured. The NDS preserves detailed operational records. For analytical purposes, selected data are then extracted and reorganized into a Dimensional Data Store (DDS). The DDS structures information according to the core indices of the MILP model—items, FGPs, and time periods—thereby enabling direct mapping between stored data and model parameters.

Decision Support System Layer

Based on the DDS, the DSS performs model instantiation by translating structured data into a fully specified MILP model. Decision variables, constraints, and cost parameters are initialized in accordance with the mathematical formulation. The instantiated model

is then solved using a MILP solver to generate optimal allocation decisions under capacity and inventory constraints. The resulting outputs, including decision variables and KPIs, are transferred to a visualization and dashboard module for managerial evaluation. The same structure also enables scenario adjustments, allowing planners to modify key parameters and re-execute the model to conduct what-if and sensitivity analyses.

User Interaction and Feedback Loop

Through the dashboard interface, decision makers can review baseline results, conduct what-if and sensitivity analyses, and either accept the solution or trigger re-solving with adjusted inputs. This establishes a controlled feedback loop linking user judgment with model re-execution, without bypassing the underlying optimization logic.

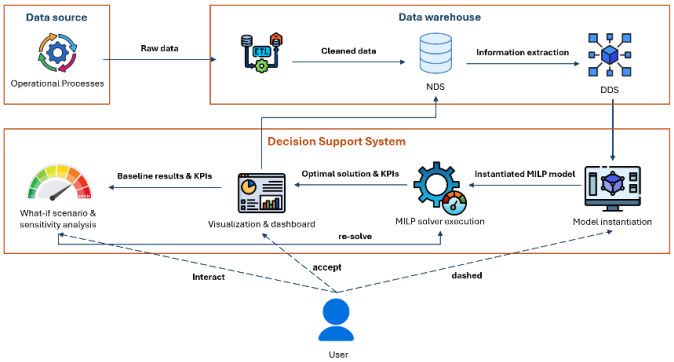

Fig. 2. DSS architecture

5 Experiment and evaluation

5.1 Case study background

The case study examines a global automotive corporation implementing an SMI structure in its supply chain. The system involves a Vietnam-based CS that manufactures seat trim cover kits and US-based FGPs where the kits are used for vehicle seat assembly. Through real-time data sharing, replenishment decisions are aligned with vehicle assembly demand and capacity constraints. The CS relies on rolling forecasts from the FGPs and plans several weeks ahead of their operational schedule. As the shipment lead time from the CS to FGP is assumed to be constant, it can be ignored to allow supplier decisions to be interpreted as preceding FGP demand realization with zero effective lead time.

This section examines a numerical example parameterized with six FGPs and a four-week planning horizon. The product portfolio includes 943 distinct seat trim cover items, which are supplied to six US-based FGPs. This scale reflects the operational complexity of managing supplier-driven replenishment decisions across multiple

products and geographically dispersed assembly plants within a global automotive supply chain.

5.2 Problem instances

To evaluate the performance of the proposed optimization model and its associated DSS for SMI, three representative problem instances are constructed to reflect distinct operational conditions, each comprising three warehouses and four planning periods. These instances progressively stress the system, ranging from demand–supply imbalance to inventory saturation and non-optimal case-pack allocation. The objective is to assess the robustness and decision-support effectiveness of the proposed framework under varying levels of operational complexity. All test cases are examined under a common set of baseline parameters as Table 3.

Table 3. Fixed parameters

Inventory floor	2
Inventory ceiling	25
Backorder unit cost	200
Overstock unit cost	2
Shortage unit cost	5
Penalty cost	100

Backorder existance

To examine the allocation behavior under severe demand imbalance, this scenario introduces a substantial negative demand shock at WH01 in the first period while maintaining limited production capacity (Table 4). The result indicates that allocation decisions are strongly driven by backorder mitigation. Under severe demand imbalance at WH01, the limited production capacity is primarily allocated to this warehouse in the early periods to reduce accumulated backorders. This behavior reflects the dominance of backorder cost in the objective function, leading the model to prioritize recovery at the most critically constrained location. Consequently, warehouses with higher shortage pressure receive priority in order allocation.

Table 4. Test instance 1: Backorder existance

Capacity = 12			Period			
Warehouse	Data Type	Case-pack	1	2	3	4
WH01	Beginning inventory	4	10			
	Inventory fluctuation	4	-40	-2	-2	-2
	Net inventory (before)	4	-30	-32	-34	-36
	Pack quantity	4	3	3	3	1
	Residual	4	0	0	0	0
	Net inventory (after)	4	-18	-8	2	4
WH02	Beginning inventory	4	15			
	Inventory fluctuation	4	-3	-3	-3	-3

	Net inventory (before)	4	12	9	6	3
	Pack quantity	4	0	0	0	1
	Residual	4	0	0	0	0
	Net inventory (after)	4	12	9	6	7
WH03	Beginning inventory	4	12			
	Inventory fluctuation	4	-3	-3	-3	-3
	Net inventory (before)	4	9	6	3	0
	Pack quantity	4	0	0	0	1
	Residual	4	0	0	0	0
	Net inventory (after)	4	9	6	3	4

Overstock penalty

In the overstock penalty scenario, inventory levels at WH01 and WH02 consistently exceed the prescribed ceiling, creating sustained overstock pressure (Table 5). The results show that order allocation is concentrated exclusively on WH03, while WH01 and WH02 receive no additional supply due to their persistent inventory levels above the ceiling. Given the overstock penalty included in the objective function, the model avoids allocating capacity to warehouses already exceeding the upper bound. Instead, the limited capacity is directed to the only warehouse facing inventory depletion, thereby mitigating shortage risk without aggravating overstock cost. This behavior confirms that allocation decisions are governed by inventory bound violations and associated penalty structures.

Table 5. Test instance 2: Overstock penalty

Capacity = 8			Period			
Warehouse	Data Type	Case-pack	1	2	3	4
WH01	Beginning inventory	4	35			
	Inventory fluctuation	4	0	0	0	0
	Net inventory (before)	4	35	35	35	35
	Pack quantity	4	0	0	0	0
	Residual	4	0	0	0	0
	Net inventory (after)	4	35	35	35	35
WH02	Beginning inventory	4	32			
	Inventory fluctuation	4	0	0	0	0
	Net inventory (before)	4	32	32	32	32
	Pack quantity	4	0	0	0	0
	Residual	4	0	0	0	0
	Net inventory (after)	4	32	32	32	32
WH03	Beginning inventory	4	12			
	Inventory fluctuation	4	-5	-5	-5	-5
	Net inventory (before)	4	7	2	-7	-12
	Pack quantity	4	2	2	2	2
	Residual	4	0	0	0	0
	Net inventory (after)	4	15	18	21	24

Non-optimal case-pack allocation

In the non-optimal case-pack allocation scenario, the production capacity of 13 units per period is not divisible by the case-pack size of 4, inherently generating residual quantities (Table 6). The results indicate that, in each period, only one warehouse incurs a non-zero residual unit, while the remaining allocations are structured as full case-packs. This pattern reflects the model's effort to confine penalty costs to a single location rather than distributing inefficiencies across multiple warehouses. By concentrating the indivisible remainder, the allocation minimizes the total penalty impact while preserving overall capacity feasibility.

Table 6. Non-optimal case-pack allocation

Capacity = 13			Period			
Warehouse	Data Type	Case-pack	1	2	3	4
WH01	Beginning inventory	4	10			
	Inventory fluctution	4	-4	-4	-4	-4
	Net inventory (before)	4	6	2	-2	-6
	Pack quantity	4	0	3	1	3
	Residual	4	1	1	0	0
	Net inventory (after)	4	7	16	16	24
WH02	Beginning inventory	4	10			
	Inventory fluctuation	4	-4	-4	-4	-4
	Net inventory (before)	4	6	2	-2	-6
	Pack quantity	4	0	0	2	0
	Residual	4	0	0	0	1
	Net inventory (after)	4	6	2	6	3
WH03	Beginning inventory	4	10			
	Inventory fluctuation	4	-4	-4	-4	-4
	Net inventory (before)	4	6	2	-2	-6
	Pack quantity	4	3	0	0	0
	Residual	4	0	0	1	0
	Net inventory (after)	4	18	14	11	7

5.3 Numerical Result

The mathematical model is solved using Python 3.14.0 on a PC equipped with an AMD Ryzen 7 5800U processor (1.90 GHz) and 16.0 GB of RAM, running a 64-bit operating system. The MILP formulation is implemented in PuLP 2.9.0 and solved with the CBC (Coin-OR Branch and Cut) solver. Default solver settings are adopted, with console output disabled (msg = False) and a maximum time limit of 30 seconds imposed for each item-level subproblem. All test instances are solved to proven optimality with zero optimality gap.

Computational Performance on Real Data

The computational results confirm the practical tractability of the proposed MILP framework at real operational scale. Across 943 item-level subproblems, the average solving time is 0.05 seconds, with a minimum of 0.01 seconds and a maximum of 0.66 seconds (Fig 3). The entire batch is completed within 51 seconds, and all instances are solved to proven optimality with zero optimality gap, without any infeasible or timeout cases. These findings demonstrate that the optimization engine can be reliably integrated into routine planning processes without introducing computational bottlenecks.

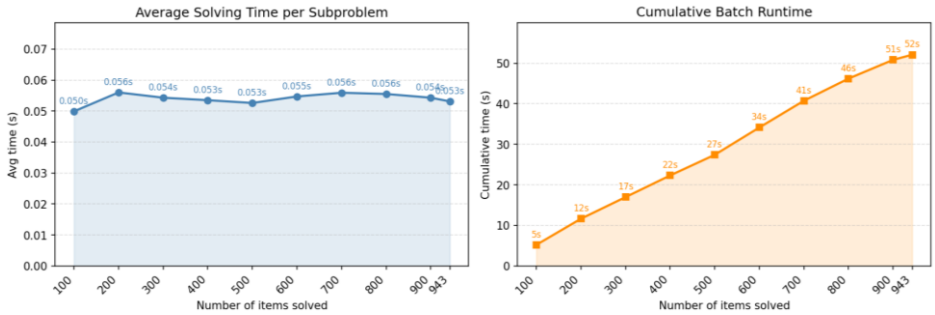

Fig. 3. Computational Scalability of the Proposed MILP Framework

Cost Structure and Allocation Behavior

To assess the economic impact of the proposed optimization framework, the optimized solution is compared with the baseline allocation policy currently used in practice. Under the baseline, total system cost reaches 135,035,759.40, whereas the optimized solution reduces it to 6,203,728.00, generating savings of 128,832,031.40 (95.41%). This result highlights the substantial economic benefit of coordinated capacity allocation within the SS-MB-SMI framework.

Aggregated results reveal several structural insights. Across 943 item-level problems, the average objective value per item is 6,578.71. Backorder cost accounts for 98.2% of total cost, although only 3.8% of items experience backorders, indicating that a few severe shortages drive most economic impact. Overstock costs occur more frequently but contribute only 1.3% to total cost, and no penalty costs are observed under the optimized policy.

From an allocation perspective, capacity decisions exhibit consistent structural patterns. In 71.6% of periods, more than 70% of available capacity is allocated to a single warehouse, reflecting targeted mitigation of critical inventory imbalances rather than uniform distribution. Residual case-pack violations do not occur, and backorders arise in only 1.4% of periods. These findings confirm that the model generates economically efficient and operationally stable allocation decisions under real-world conditions.

5.4 Managerial insights

The findings of this study provide several strategic implications for managerial decision-making in vertically integrated SMI systems.

First, production capacity flexibility emerges as the primary structural lever of system performance. The numerical results indicate that a small number of severe backorder cases account for most of the total system cost. Consequently, incremental improvements in capacity flexibility—such as temporary capacity expansion, overtime policies, or improved reallocation rules—can yield disproportionate economic benefits compared to marginal adjustments in routine replenishment quantities.

Second, inventory ceiling and floor parameters function as policy instruments that shape allocation priorities and service outcomes. Rather than being treated as fixed operational constraints, these bounds should be periodically reviewed and aligned with strategic service targets. Adjustments to safety stock policy can significantly alter the trade-off between service reliability and inventory-related costs at the system level.

Third, the DSS framework enhances managerial visibility of cost decomposition and allocation behavior, enabling transparent evaluation of trade-offs among backorder, overstock, and case-pack inefficiencies. This visibility supports evidence-based investment decisions, particularly when choosing between expanding storage capacity and improving production flexibility.

Overall, the results suggest that strategic investment in capacity responsiveness and decision-support capabilities has a greater long-term impact on supply chain performance than isolated adjustments in warehouse inventory levels.

6 Conclusion

This study develops a MILP model to address the SS-MB-SMI problem by optimizing order allocation from a single component supplier to multiple finished-goods plants. To support practical implementation, a DSS is designed to integrate data input, model execution, and automated solution generation within an interactive platform. A case study in the automotive industry demonstrates the model's ability to enhance coordinated decision-making, improving allocation efficiency, capacity utilization, and inventory control across the supply chain.

Two main limitations are identified. First, all model parameters are deterministic, which does not fully capture real-world uncertainty such as demand volatility, supply disruptions, or operational variability. Second, the framework assumes fully rational decision-making, while actual managerial decisions may reflect risk preferences or organizational constraints.

Future research should incorporate uncertainty through stochastic or robust optimization approaches and extend the DSS to account for behavioral considerations, such as risk attitudes or multi-objective preferences. These extensions would enhance the realism and managerial relevance of the proposed framework.

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